PulseTwin

SYSTEM ARCHITECTURE / IMPLEMENTED PROTOTYPE

A transparent observation-to-scenario loop.

01 Browser

HTML / CSS / JavaScript

Controls, SVG, CSV, JSON export

→

02 API contract

FastAPI / Pydantic

Schema, bounds, consecutive dates

→

03 Personal model

NumPy / Bayesian + Kalman

Fit, state, forecast, holdout

REQUEST

Observations + hypothetical inputs + horizon -> validated request -> personal fit and state -> conditional trajectory + evidence

TRUST BOUNDARY

Local browser and server only. No database, authentication, live devices or external AI service. Reports contain input data. Use synthetic or authorized de-identified records.

MODEL FLOW / EVALUATION SEPARATION

One history. Two separate model fits.

Scenario path

All supplied history -> Bayesian coefficient posterior

Residual sequence -> Kalman state

Recent 7-day habits -> assumed gradual scenario

Mean + 95% predictive band -> contribution report

Evaluation path

Chronological training prefix only -> separate fit

Held-out daily habits -> conditional predictions

Held-out BP labels -> MAE and interval coverage

Last reading + training mean -> comparison baselines

Important: no held-out BP label enters the evaluation fit. Known held-out habits make this conditional prediction, not prospective causal validation.